

The Strategic Mass-Management Problem

Celestial Risk, Resource Conversion, Hierarchical Compression, Planetary Defense, and the Optimal Management of Solar-System Mass

Lord Soumadeep Ghosh with ChatGPT (OpenAI)

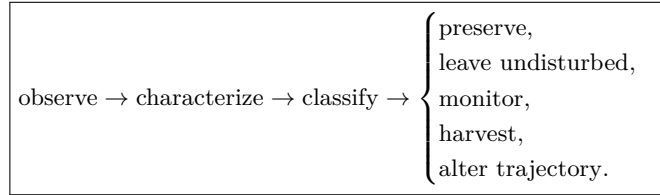
Kolkata, India

Abstract

This paper introduces the *Strategic Mass-Management Problem*: the problem of determining which celestial masses should remain autonomous natural bodies, which should be preserved, which should be observed, which may eventually be harvested as resources, and which hazardous trajectories should be modified, subject to physical conservation laws, technological feasibility, uncertainty, intervention risk, economic cost, strategic stability, and intergenerational welfare.

The framework departs from an unnecessarily expansive interpretation of the many-body problem. Discovery of another celestial object does not imply that the object must become another independent strategic unit. Instead, celestial masses may be classified according to their dynamical importance, resource value, hazard, scientific value, ecological or civilizational importance, and controllability.

The basic policy architecture is



A crucial distinction is made between *mass destruction* and *mass transformation*. Celestial resource utilization does not remove mass from the physical system. Rather,

$$m_{\text{natural}} \rightarrow m_{\text{infrastructure}} + m_{\text{inventory}} + m_{\text{residual}} + m_{\text{waste}},$$

subject to conservation laws. Hence the strategic objective is not the elimination of mass but the transformation of some uncontrolled natural capital into controlled physical and economic capital when doing so raises expected welfare.

The framework integrates celestial mechanics, astronomy, astrophysics, cosmology, planetary defense, physics, chemistry, materials science, biology, ecology, economics, finance, welfare economics, political science, international relations, geopolitics, warfare economics, decision theory, statistics, computer science, control theory, operations research, data science, artificial intelligence, machine learning, network science, psychology, psychiatry, philosophy, ethics, and intergenerational choice.

The principal conclusion is conservative. Harvesting is not automatically risk reduction, fragmentation is not automatically safety, and proximity to an asteroid population is not by itself a sufficient measure of hazard. Optimal celestial mass management requires observation, probabilistic orbit determination, counterfactual simulation, robust decision rules, conservation constraints, and explicit comparison of the risks of action with the risks of inaction.

The paper ends with “The End”

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1 Introduction

A conventional expansion of an N -body model proceeds by adding bodies:

$$N \rightarrow N + 1 \rightarrow N + 2 \rightarrow \cdots .$$

This is mathematically natural when every newly introduced object materially affects the dynamics under investigation.

It is not necessarily strategically optimal.

Suppose a civilization cares primarily about a small number of inhabited, economically useful, scientifically valuable, or dynamically important celestial bodies. The discovery of an additional asteroid does not imply that the asteroid should be treated as another strategic peer.

Instead, one may ask:

What should be done with the mass represented by the newly considered object?

This produces a different problem.

For each celestial object i , choose a policy

$$a_i \in \{P, L, M, H, D\},$$

where

- P denotes preservation,
- L denotes leaving the object undisturbed,
- M denotes monitoring,
- H denotes prospective harvesting,
- D denotes trajectory modification or deflection when justified.

The resulting problem is the *Strategic Mass-Management Problem*.

2 From Body Management to Mass Management

Let the Solar-System environment contain bodies

$$\mathcal{B} = \{B_1, \dots, B_N\}.$$

Body B_i possesses mass m_i , state

$$X_i(t) = (\mathbf{r}_i, \mathbf{v}_i, m_i, \chi_i),$$

where χ_i denotes relevant composition and structural characteristics.

A body-oriented framework asks

How does B_i move?

A strategic mass-management framework asks the broader question

What role should the mass m_i play in the future system?

Possible answers include

natural body,

scientific reserve,

monitored hazard,

resource stock,

constructed infrastructure,

or combinations thereof.

3 The Non-Addition Principle

Principle 3.1 (Non-Addition Principle). *Discovery or identification of an additional physical body does not by itself imply that the body should be promoted to an independent strategic unit.*

Symbolically,

$$N_B \uparrow \not\Rightarrow N_{\text{strategic}} \uparrow.$$

The object should become an independent strategic unit only if its dynamical, informational, economic, political, scientific, or security importance requires such treatment.

This separates

physical multiplicity

from

strategic multiplicity.

4 A Strategic Classification of Celestial Mass

For body i , define the state vector

$$Z_i = (H_i, R_i, S_i, C_i, D_i, U_i, Q_i),$$

where

- H_i = hazard,
- R_i = recoverable resource value,
- S_i = scientific value,
- C_i = intervention cost,
- D_i = intervention-induced dynamical risk,
- U_i = uncertainty,
- Q_i = strategic or civilizational value.

A classification rule is

$$a_i = \pi(Z_i).$$

The policy π should not be determined by hazard alone.

A resource-rich object may be too dangerous to disturb.

A harmless object may have exceptional scientific value and should be preserved.

A potentially hazardous object may require only observation.

A threatening object may require trajectory modification rather than resource extraction.

5 Mass Cannot Be Eliminated

The phrase “harvesting risky mass” must be interpreted physically.

Mass is not simply deleted.

For a resource body of initial mass m_i , write

$$m_i = m_i^I + m_i^F + m_i^W + m_i^R,$$

where

- m_i^I = infrastructure mass,
- m_i^F = feedstock or inventory,
- m_i^W = waste or discarded material,
- m_i^R = residual natural body.

Thus

$$m_i \rightarrow \{m_i^I, m_i^F, m_i^W, m_i^R\}.$$

The strategic transformation is therefore

$$\text{uncontrolled natural mass} \rightarrow \text{managed distribution of mass.}$$

6 Conservation of Momentum

Mass transformation is constrained by momentum conservation.

For an isolated local interaction,

$$\sum_i m_i \mathbf{v}_i = \mathbf{P}.$$

If material of mass Δm is removed from an asteroid with ejection velocity $\mathbf{u}$, the residual body receives a momentum change.

Schematically,

$$m\mathbf{v} = (m - \Delta m)(\mathbf{v} + \Delta\mathbf{v}) + \Delta m(\mathbf{v} + \mathbf{u}).$$

To first order,

$$m\Delta\mathbf{v} \approx -\Delta m\mathbf{u}.$$

Therefore

$$\Delta\mathbf{v} \approx -\frac{\Delta m}{m}\mathbf{u}.$$

Resource extraction is consequently also an orbital perturbation problem.

7 Harvesting Is Not Automatically Risk Reduction

This point is fundamental.

Suppose an asteroid has baseline hazard H_0 .

An intervention produces residual hazard

$$H_1 = H_0 - \Delta H_{\text{benefit}} + H_{\text{intervention}}.$$

Harvesting improves safety only if

$$H_1 < H_0.$$

Equivalently,

$$\Delta H_{\text{benefit}} > H_{\text{intervention}}.$$

Intervention risk may include

- orbit-estimation error,
- unintended momentum transfer,
- fragmentation,
- poorly controlled ejecta,
- spacecraft failure,
- incomplete extraction,
- uncertain material properties,
- long-horizon orbital instability.

Hence “mine the dangerous asteroid” is not a general solution.

8 Fragmentation Risk

Suppose one body of mass M fragments into n bodies:

$$M = \sum_{j=1}^n m_j.$$

Before fragmentation, the hazard is

$$H(M, X).$$

After fragmentation,

$$H' = \mathcal{F}(m_1, \dots, m_n, X_1, \dots, X_n).$$

There is no theorem implying

$$H' < H.$$

Indeed, the number of trajectories requiring prediction increases from

$$1$$

to

$$n.$$

Therefore fragmentation may reduce, preserve, or increase strategic risk.

9 The Safe-Intervention Principle

Principle 9.1 (Safe-Intervention Principle). *A celestial intervention should not be undertaken merely because an object is potentially hazardous or economically valuable. Intervention should occur only when the expected risk-adjusted benefit exceeds the expected risk-adjusted value of non-intervention.*

Let

$$V(a_i \mid Z_i)$$

be expected welfare from action a_i .

Then

$$a_i^* = \arg \max_{a_i \in \mathcal{A}} V(a_i \mid Z_i).$$

In particular, harvesting is optimal only if

$$V(H \mid Z_i) > \max\{V(P \mid Z_i), V(L \mid Z_i), V(M \mid Z_i), V(D \mid Z_i)\}.$$

10 Planetary Hazard Is a Flow Problem

The risk associated with asteroids is not determined merely by geographic proximity to the main asteroid belt.

Gravitational perturbations, resonances, encounters, and secular evolution can move small bodies through orbital-element space.

Let

$$\mathbf{y}_i = (a_i, e_i, I_i, \Omega_i, \omega_i, M_i)$$

be the orbital elements of body i .

Then

$$\dot{\mathbf{y}}_i = F_i(\mathbf{y}_1, \dots, \mathbf{y}_N, t).$$

The relevant hazard is therefore associated with dynamical flows

$$\mathbf{y}_i(t_0) \rightarrow \mathbf{y}_i(t).$$

Hence

$$\boxed{\text{proximity to asteroid belt} \neq \text{complete collision-risk measure.}}$$

11 A Spatial Hazard Functional

Let $\mathbf{r}$ be a location of strategic interest.

Define

$$\boxed{\mathcal{H}(\mathbf{r}, t) = \sum_i P_i(\mathbf{r}, t) L_i(\mathbf{r}, t),}$$

where

- $P_i(\mathbf{r}, t)$ is a modeled encounter or impact probability,
- $L_i(\mathbf{r}, t)$ is conditional loss.

A more complete functional includes uncertainty:

$$\mathcal{H} = \mathbb{E}[L] + \lambda \text{Var}(L) + \eta U,$$

where U measures model uncertainty.

This permits comparisons among Earth, Moon, Mars, orbital habitats, and other locations without reducing the problem to distance from the asteroid belt.

12 Mars as a Backup Planet

Mars may possess strategic value independently of its resource value.

Let

$$Q_M = Q_M^{\text{hab}} + Q_M^{\text{science}} + Q_M^{\text{redundancy}} + Q_M^{\text{resources}}.$$

The redundancy term is particularly important.

Suppose a civilization exists on Earth alone. Let catastrophic survival probability be

$$P_E.$$

If an independently survivable Martian settlement exists, the probability that at least one settlement survives may be

$$P_{\text{survival}} = 1 - (1 - P_E)(1 - P_M)$$

under an illustrative independence assumption.

Correlation modifies this expression.

If failures are strongly correlated, the diversification benefit falls.

Thus the strategic value of Mars depends not merely on inhabitation but on the degree to which it creates genuinely independent civilizational redundancy.

13 Portfolio Theory of Planetary Redundancy

Civilizational locations can be treated abstractly as a portfolio.

Let

$$\mathbf{w} = (w_E, w_M, w_L, \dots)$$

denote fractions of strategic capital allocated among locations.

Let expected welfare returns be $\boldsymbol{\mu}$ and systemic covariance be Σ .

Then a mean-risk objective is

$$\max_{\mathbf{w}} \left[\mathbf{w}^T \boldsymbol{\mu} - \frac{\gamma}{2} \mathbf{w}^T \Sigma \mathbf{w} \right].$$

The analogy is limited but useful.

A “backup planet” has greatest diversification value when its catastrophic risk is not perfectly correlated with Earth’s.

14 Jupiter as a Systemically Important Mass

Not all celestial masses should be treated as harvestable or subordinate objects.

A massive body may be dynamically systemically important.

Define the characteristic perturbation ratio

$$\Gamma_{ij} = \frac{a_{j \rightarrow i}^{\text{pert}}}{a_i^{\text{binding}}}.$$

If

$$\Gamma_{ij} \ll 1,$$

body j may be treated as a weak perturbation over an appropriate horizon.

If

$$\Gamma_{ij} \not\ll 1,$$

that approximation may fail.

Jupiter’s large mass makes its dynamical influence fundamentally different from that of a small asteroid.

Thus the theory distinguishes

$$\text{resource-scale mass}$$

from

$$\text{systemically important mass.}$$

15 The Systemic-Mass Criterion

Definition 15.1 (Systemically Important Celestial Body). *A celestial body is systemically important relative to a target system, time horizon, and tolerance if neglecting or coarsely compressing its gravitational influence produces prediction errors larger than the permitted tolerance.*

Let

$$X_{\text{full}}(t)$$

be the reference dynamics and

$$X_{-j}(t)$$

the dynamics under a model in which body j is neglected or excessively compressed.

Define

$$E_j(T) = \sup_{0 \leq t \leq T} \|X_{\text{full}}(t) - X_{-j}(t)\|.$$

Then body j is systemically important at tolerance ϵ if

$$E_j(T) > \epsilon.$$

This gives an operational rather than rhetorical definition of “important mass.”

16 Hierarchical Compression Boundary

The preceding argument motivates a central theoretical problem.

Definition 16.1 (Hierarchical Compression Boundary). *The hierarchical compression boundary is the set of parameter values beyond which a celestial body or subsystem can no longer be represented as a subordinate or compressed component without violating a specified prediction, control, welfare, or risk tolerance.*

If

$$E_j(T) \leq \epsilon,$$

compression is acceptable.

If

$$E_j(T) > \epsilon,$$

the object must be promoted to a more explicit level of the model.

Hence

$$\text{compression} \leftrightarrow \text{error tolerance.}$$

17 The Strategic Mass-Management Objective

Let policy π determine the treatment of all manageable objects.

Define

$$J(\pi) = \mathbb{E} \int_0^T e^{-\rho t} [L_H(t) + \lambda_I L_I(t) + \lambda_C C(t) + \lambda_U U(t) - \lambda_R R(t) - \lambda_S S(t)] dt,$$

where

- L_H = natural celestial hazard,
- L_I = intervention-generated hazard,
- C = economic cost,
- U = uncertainty penalty,
- R = resource benefit,
- S = scientific or strategic benefit.

The planner solves

$$\pi^* = \arg \min_{\pi \in \Pi} J(\pi).$$

This is the core Strategic Mass-Management Problem.

18 Constrained Formulation

The optimization is subject to

$$\dot{X} = f(X, u),$$

$$u(t) \in \mathcal{U}(X_t),$$

$$m_{\text{before}} = m_{\text{after}} + \frac{E_{\text{released}}}{c^2}$$

when mass-energy accounting is relevant,

$$P_{\text{before}} = P_{\text{after}},$$

and technological constraints

$$g_k(X, u) \leq 0.$$

Safety constraints may require

$$\mathbb{P}(\text{catastrophic impact} \mid \pi) \leq \bar{p}.$$

Thus economic optimization is subordinate to physical feasibility.

19 A Basic Intervention Theorem

Theorem 19.1 (Intervention Dominance Criterion). *Let $a = 0$ denote non-intervention and $a = 1$ denote intervention. Suppose expected total losses are*

$$L_0 = H_0$$

under non-intervention and

$$L_1 = H_0 - \Delta H + B_I + C_I$$

under intervention, where ΔH is expected hazard reduction, B_I is expected intervention-generated loss, and C_I is intervention cost expressed in common welfare units.

Then intervention is preferred if and only if

$$\boxed{\Delta H > B_I + C_I.}$$

Proof. Intervention is preferred exactly when

$$L_1 < L_0.$$

Substituting gives

$$H_0 - \Delta H + B_I + C_I < H_0.$$

Canceling H_0 yields

$$-\Delta H + B_I + C_I < 0,$$

or

$$\Delta H > B_I + C_I.$$

□

The theorem is elementary, but its implication is important: a hazardous object does not automatically justify intervention.

20 Resource Economics

Let asteroid i contain quantities

$$q_{ik}$$

of resource k .

Let the shadow value of resource k at destination d be

$$p_{kd}.$$

Gross resource value is

$$V_i^R = \sum_{k,d} p_{kd} q_{ikd}.$$

But recoverable value must subtract

$$C_i^{\text{survey}},$$

$$C_i^{\text{capture}},$$

$$C_i^{\text{processing}},$$

$$C_i^{\text{transport}},$$

$$C_i^{\text{risk}}.$$

Therefore net value is

$$\boxed{NPV_i = V_i^R - C_i^{\text{survey}} - C_i^{\text{capture}} - C_i^{\text{processing}} - C_i^{\text{transport}} - C_i^{\text{risk}}}.$$

An asteroid can contain valuable material while having negative economic value under current technology.

21 Natural Capital to Produced Capital

Celestial resources may be represented as natural capital K_N .

Extraction transforms them into produced capital K_P :

$$K_N \rightarrow K_P.$$

More precisely,

$$\dot{K}_N = -X_t,$$

$$\dot{K}_P = \eta X_t - \delta K_P,$$

where

- X_t is extraction,
- η is conversion efficiency,
- δ is depreciation.

This creates a celestial version of the classical resource-extraction problem.

22 Hotelling-Type Resource Condition

For a non-renewable celestial resource with extraction cost c_t , let p_t denote its shadow price.

Under idealized competitive assumptions, an intertemporal efficiency condition takes the form

$$\frac{\dot{p}_t}{p_t} \approx r$$

for the scarcity component of price.

However, celestial resources introduce unusual corrections:

$$r \rightarrow r + \text{transport risk} + \text{mission risk} + \text{political risk} + \text{orbital risk}.$$

Hence extraterrestrial resource economics cannot simply import terrestrial mining formulas without modification.

23 Finance

A resource mission is a contingent claim.

Let terminal project payoff be

$$X_T.$$

Its value is

$$V_t = \mathbb{E}_t[m_{t,T} X_T],$$

where $m_{t,T}$ is a stochastic discount factor.

Write

$$X_T = R_T - C_T - L_T,$$

where R_T is resource revenue, C_T cost, and L_T accident or failure loss.

Mission value is therefore

$$V_t = \mathbb{E}_t[m_{t,T}(R_T - C_T - L_T)].$$

A positive geological resource estimate does not imply positive financial value.

24 Real Options

Waiting has value when technology is improving and uncertainty is high.

Let $V_i(t)$ be the value of harvesting object i and $K_i(t)$ the investment cost.

Immediate exercise gives

$$V_i(t) - K_i(t).$$

But the option to wait has value

$$F_i(t).$$

Therefore extraction should occur only when

$$V_i(t) - K_i(t) \geq F_i(t).$$

This is especially relevant for celestial resources because technological learning may substantially lower future extraction costs.

25 Welfare Economics

A social planner chooses celestial policy to maximize

$$W = \mathbb{E} \int_0^\infty e^{-\rho t} \left[U(C_t) - L_t + B_t^{\text{science}} + B_t^{\text{security}} \right] dt.$$

The planner faces an intergenerational problem.

Extracting a unique object today may destroy scientific information or future option value.

Conversely, failure to mitigate a genuine hazard can impose catastrophic losses on future generations.

Thus the relevant welfare comparison includes

$$\boxed{\text{present benefit} + \text{future option value} + \text{risk reduction} - \text{irreversible loss.}}$$

26 Precaution under Knightian Uncertainty

Suppose the true probability distribution is not known.

Let

$$P \in \mathcal{P}$$

belong to an ambiguity set.

A robust policy solves

$$\boxed{\pi^* = \arg \max_{\pi} \min_{P \in \mathcal{P}} \mathbb{E}_P[W(\pi)].}$$

This can justify conservative policies when the downside of intervention is large and poorly characterized.

The existence of uncertainty therefore does not imply paralysis.

It implies that uncertainty itself enters the optimization problem.

27 Planetary Defense

Planetary defense can be represented as a sequence:

$$\boxed{\text{detect} \rightarrow \text{track} \rightarrow \text{estimate orbit} \rightarrow \text{estimate impact probability} \rightarrow \text{characterize} \rightarrow \text{select response.}}$$

Possible response classes include

observation,

civil protection,

trajectory modification,

and, in a future technological regime,

controlled resource utilization.

The response must be matched to the object and warning time.

28 Small Changes and Long Horizons

For orbital hazard reduction, large instantaneous changes are not always necessary.

Suppose a small velocity change Δv is applied sufficiently early.

Over a long time T , a schematic displacement scale is

$$\Delta x \sim T \Delta v,$$

although actual orbital dynamics require propagation through the gravitational system.

Thus early detection has strategic value because

warning time

can substitute partially for

control magnitude.

This is one reason observation infrastructure may have exceptionally high social value.

29 Astronomy as Preventive Infrastructure

Let survey expenditure be A_t .

Suppose improved observation lowers posterior orbital uncertainty:

$$\frac{\partial U_t}{\partial A_t} < 0.$$

If intervention cost increases with uncertainty,

$$\frac{\partial C_I}{\partial U} > 0,$$

then

$$\frac{\partial C_I}{\partial A} < 0.$$

Astronomical observation therefore possesses an economic value beyond pure science.

It creates an information capital stock.

30 Information Capital

Let

$$K_t^I$$

denote astronomical information capital.

Then

$$\dot{K}_t^I = A_t - \delta_I K_t^I.$$

Higher information capital improves

orbit estimation,

composition estimation,

mission planning,

risk assessment.

Thus the strategic production function may be written

$$Y_t = F(K_t, L_t, K_t^I, R_t).$$

Knowledge becomes an economically productive input.

31 Physics of Resource Extraction

For a body of radius R and mass M , the escape speed is

$$v_{\text{esc}} = \sqrt{\frac{2GM}{R}}.$$

The characteristic gravitational energy required to remove a unit mass from its surface is of order

$$\epsilon_g \sim \frac{GM}{R}.$$

Small bodies can therefore offer energetic advantages relative to massive planets.
But low gravity also means that excavation products can escape readily.
Hence

low gravity = opportunity + containment problem.
--

32 Chemistry and Mineralogy

Resource value depends upon composition.

Let the composition vector be

$$\mathbf{q}_i = (q_{i1}, q_{i2}, \dots, q_{ik}).$$

Chemical processing can be represented by

$$\dot{\mathbf{n}} = S\mathbf{v},$$

where S is a stoichiometric matrix and $\mathbf{v}$ is a vector of reaction rates.
Energy requirements satisfy

$$E_{\text{process}} = \sum_j \Delta H_j \xi_j + E_{\text{separation}} + E_{\text{transport}}.$$

Thus “resource abundance” is not equivalent to “usable resource abundance.”
Extractability and processing matter.

33 Materials Science

A target body may be

monolithic,

fractured,

porous,

rubble-pile-like.

Mechanical response depends upon quantities such as

$$\sigma_y, \quad E_Y, \quad \nu, \quad \phi,$$

representing yield strength, Young’s modulus, Poisson ratio, and porosity.

Uncertainty in these quantities affects both mining feasibility and deflection response.
Consequently,

$$\boxed{\text{orbital state estimation} + \text{material characterization}}$$

are complementary requirements.

34 Biology and Life-Support Constraints

If harvested mass supports settlements, material value is partly determined by biological requirements.
Let stocks include

$$W_t = \text{water},$$

$$O_t = \text{oxygen},$$

$$N_t = \text{nutrient inventory}.$$

A closed-loop life-support system may be represented as

$$\dot{\mathbf{z}} = F(\mathbf{z}, \mathbf{u}, \mathbf{r}),$$

where $\mathbf{r}$ denotes externally supplied resources.

Celestial material therefore has value not merely as industrial feedstock but potentially as part of the metabolism of off-world settlements.

35 Ecology and Planetary Protection

Economic value is not the only relevant value.

If an object has scientific importance for understanding prebiotic chemistry or the history of the Solar System, extraction can impose irreversible knowledge loss.

Let

$$S_i$$

denote scientific preservation value.

Then the harvesting condition should include

$$V_i^R - C_i - S_i > 0$$

only as a simplified first approximation.

If there is credible biological or contamination significance, additional planetary-protection constraints are required.

36 Psychology of Rare Catastrophic Risk

Humans may systematically misperceive low-probability catastrophic events.

Let objective probability be p .

A behavioral decision weight may be

$$w(p) \neq p.$$

For example, some decision makers may overweight salient rare events, while others may neglect distant risks.

Therefore policy should distinguish

$$\boxed{\text{estimated physical probability}}$$

from

psychological probability weighting.

Scientific risk assessment should not be replaced by fear or complacency.

37 Psychiatric Caution

Concern about celestial catastrophe does not itself imply psychiatric pathology.

Likewise, indifference to low-probability hazards does not establish a clinical condition.

Abstract models such as

$$q_{t+1} = (1 - \alpha)q_t + \alpha y_t$$

describe belief updating.

They should not be interpreted as psychiatric diagnoses without independent clinical evidence.

This distinction is necessary whenever strategic-risk theory incorporates psychological or psychiatric terminology.

38 Philosophy of Intervention

The Strategic Mass-Management Problem raises a philosophical question:

When does knowledge of nature justify deliberate transformation of nature?

There are at least four positions:

1. preserve natural celestial objects whenever possible;
2. transform them whenever expected economic value is positive;
3. intervene primarily to reduce serious hazard;
4. optimize across preservation, science, resources, security, and future option value.

The fourth is adopted here.

It avoids both

unconditional exploitation

and

unconditional non-intervention.

39 Ethics and Irreversibility

Suppose an action is irreversible.

Let V_{future} denote future option value.

Then immediate exploitation is justified only if

$$V_{\text{now}} > V_{\text{future}} + V_{\text{preservation}} + V_{\text{information}} + V_{\text{risk}}.$$

Because future technological capabilities may exceed present capabilities, irreversibility creates an additional reason to value delay.

40 Political Science

Celestial resources create a collective-action problem.

Suppose states $j = 1, \dots, n$ choose extraction levels x_j .

Total extraction is

$$X = \sum_j x_j.$$

Each state receives benefit

$$B_j(x_j)$$

but intervention risk may be shared:

$$C_j = C_j(X).$$

Then individual incentives can produce

$$X^{NE} > X^{SO},$$

where NE denotes a non-cooperative equilibrium and SO denotes a social optimum.

This creates a possible celestial commons problem.

41 International Relations

Celestial mass management involves questions of

- observation rights,
- information sharing,
- resource claims,
- liability,
- mission coordination,
- planetary defense,
- scientific preservation,
- dual-use technology,
- dispute resolution.

A cooperative objective is

$$\max_{\pi} \sum_j \omega_j W_j(\pi).$$

But decentralized states maximize

$$W_j(\pi_j, \pi_{-j}).$$

The difference creates a strategic game.

42 Geopolitics

Celestial infrastructure can create strategic asymmetry.

Let state j possess capabilities

$$K_j = (K_j^{\text{launch}}, K_j^{\text{survey}}, K_j^{\text{robotics}}, K_j^{\text{processing}}, K_j^{\text{navigation}}).$$

The distribution of these capabilities affects bargaining power.

Thus celestial resource policy cannot be analyzed solely through engineering or economics.

Institutional design matters.

43 Warfare Economics

Space infrastructure can be dual-use.

Let a government allocate resources

$$Y = C + I + A + D,$$

where

- C = civilian consumption,
- I = productive investment,
- A = astronomical and space infrastructure,
- D = defense.

Some space investment may contribute simultaneously to

A

and

D .

This creates accounting and governance problems.

The present framework emphasizes planetary defense and peaceful resource management rather than operational weaponization.

44 Strategic Stability

A technology capable of altering celestial trajectories may generate strategic concern even when developed for peaceful purposes.

Let state i observe action a_j but not perfectly observe intention θ_j .

Then

$$P(\theta_j \mid a_j) \propto P(a_j \mid \theta_j)P(\theta_j).$$

Incomplete information can create misperception.

Therefore transparency, verification, and international observation may have strategic value.

45 Cosmological Perspective

The Strategic Mass-Management Problem is local relative to cosmological scales.

Write physical distance as

$$\mathbf{r} = a(t)\mathbf{x}.$$

For strongly bound Solar-System structures, local gravitational dynamics dominate cosmological expansion.

Nevertheless, cosmology supplies an important conceptual lesson:

matter is organized hierarchically.

Symbolically,

small body $\subset$ planetary system $\subset$ stellar system $\subset$ galaxy $\subset$ cosmic structure.

Mass management is therefore inherently scale-dependent.

46 Statistics of Orbital Risk

Let orbital state satisfy

$$X_t \sim p(X_t \mid Y_{1:t}).$$

Impact probability over horizon T is

$$P_{\text{impact}} = \int \mathbf{1}_{\text{impact}}(X) p(X \mid Y) \, dX.$$

Expected loss is

$$\mathbb{E}[L] = \int L(X) p(X \mid Y) \, dX.$$

The policy should respond to the posterior distribution rather than merely to a point estimate of the orbit.

47 Bayesian Updating

As new observations arrive,

$$p(X \mid Y_{1:t}) \propto p(Y_t \mid X) p(X \mid Y_{1:t-1}).$$

Thus strategic classification is dynamic.

An object initially assigned to

$$M = \text{monitor}$$

may later be assigned to

$$L = \text{leave undisturbed}$$

or, if evidence changes materially,

$$D = \text{trajectory modification.}$$

Policy should therefore be adaptive.

48 Value of Information

Let optimal expected welfare without new information be

$$W_0.$$

Let Y be a prospective observation.

Expected welfare after observing Y is

$$W_1 = \mathbb{E}_Y \left[\max_a \mathbb{E}[W(a) \mid Y] \right].$$

The expected value of information is

$$\boxed{EVI = W_1 - W_0.}$$

Observation is economically justified when

$$EVI > C_{\text{observation}}.$$

For potentially hazardous objects, information may be valuable even when no physical intervention is eventually required.

49 Computer Science and State-Space Reduction

A complete Solar-System simulation can contain an enormous number of objects.

Not all objects need equal computational resolution.

Let

$$\alpha_i \in \{0, 1, 2, \dots\}$$

denote model resolution.

Then computational allocation solves

$$\min_{\alpha} \sum_i C_i(\alpha_i)$$

subject to

$$E_{\text{prediction}}(\alpha) \leq \epsilon.$$

Thus the Strategic Mass-Management Problem also becomes an adaptive computation problem.

50 Algorithms

A basic strategic classification algorithm is:

Algorithm: Strategic Celestial Mass Classification

Input:

observational data
orbital posterior
mass and composition estimates
resource estimates
scientific value
intervention technology
welfare parameters

For each object i :

1. Estimate orbital state distribution.
2. Estimate natural hazard H_i .
3. Estimate resource value R_i .
4. Estimate scientific/preservation value S_i .
5. Estimate intervention cost C_i .
6. Estimate intervention-generated risk D_i .
7. Estimate uncertainty U_i .
8. Simulate candidate policies:
 - preserve
 - leave undisturbed
 - monitor
 - harvest
 - trajectory modification
9. Compute expected welfare for each policy.

10. Apply safety and legal constraints.
11. Select the admissible policy with highest robust expected welfare.
12. Continue observation and update classification.

Output:

- policy recommendation
- posterior uncertainty
- expected welfare
- risk decomposition

51 Planetary-Defense Algorithm

Algorithm: Conservative Hazard Response

Input:

- candidate object
- observations Y
- horizon T

1. Infer posterior orbit $p(X | Y)$.
2. Propagate orbit ensemble through horizon T .
3. Estimate impact probability.
4. Estimate conditional impact loss.
5. If risk is negligible:
continue appropriate observation.
6. If uncertainty dominates:
acquire additional observations.
7. If credible hazard remains:
compare intervention strategies.
8. For every intervention:
simulate nominal outcome
simulate failure modes
simulate uncertainty
estimate secondary hazard
9. Reject interventions whose induced risk exceeds acceptable limits.
10. Select the safest admissible response maximizing expected welfare.
11. Continue observation after intervention.

52 AI and Machine Learning

Machine learning can assist in classification, but should not replace physical dynamics.

Let

$$Z_i = f_\theta(X_i, Y_i, \chi_i)$$

be a learned representation.
A policy network may estimate

$$\pi_\theta(a \mid Z_i).$$

Training loss can include

$$\mathcal{L} = \mathcal{L}_{\text{prediction}} + \lambda_1 \mathcal{L}_{\text{physics}} + \lambda_2 \mathcal{L}_{\text{risk}} + \lambda_3 \mathcal{L}_{\text{calibration}}.$$

The physics penalty can require approximately

$$m_i \ddot{\mathbf{r}}_i + G m_i \sum_{j \neq i} m_j \frac{\mathbf{r}_i - \mathbf{r}_j}{|\mathbf{r}_i - \mathbf{r}_j|^3} \approx 0.$$

This creates a physics-informed learning architecture.

53 Active Learning

Observation resources should be directed toward objects for which additional information is most valuable.

Let uncertainty reduction from observation o be

$$\Delta U(o).$$

Let cost be

$$C(o).$$

Choose

$$o^* = \arg \max_o \frac{EVI(o)}{C(o)}.$$

This is an active-learning problem.

Astronomical survey scheduling can therefore be optimized jointly with celestial risk management.

54 Reinforcement Learning

For sequential decision making, define state

$$s_t = (X_t, P_t, R_t, K_t, U_t),$$

action

$$a_t \in \mathcal{A},$$

and reward

$$r_t = B_t - C_t - L_t.$$

The objective is

$$\max_{\pi} \mathbb{E}_{\pi} \sum_{t=0}^{\infty} \gamma^t r_t.$$

For safety-critical celestial interventions, unconstrained reinforcement learning is inappropriate. Instead require

$$\mathbb{P}_{\pi}(\text{catastrophic failure}) \leq \epsilon.$$

Thus the relevant field is constrained or safe reinforcement learning.

55 Digital Twins

A strategically important object can have a continuously updated digital twin.

Let the physical object be B_i and its computational model be

$$\hat{B}_i(t).$$

Observations update

$$\hat{B}_i(t) \leftarrow \mathcal{U}(\hat{B}_i(t^-), Y_t).$$

Candidate interventions are first evaluated on the digital twin.

The cycle is

observe $\rightarrow$ update twin $\rightarrow$ simulate $\rightarrow$ decide $\rightarrow$ observe.

56 Network Science

Represent celestial bodies as nodes.

Define edge weights

$$w_{ij} = \frac{Gm_i m_j}{r_{ij}^2}.$$

Alternatively, define a normalized strategic influence measure

$$\tilde{w}_{ij} = \frac{a_{j \rightarrow i}}{a_i^{\text{reference}}}.$$

Bodies with large network centrality cannot be treated like negligible small bodies merely because they are geographically distant.

This provides a network interpretation of systemic celestial importance.

57 Control Theory

Suppose an intervention changes a body's dynamics according to

$$\dot{X} = f(X) + B(X)u.$$

The controller seeks to minimize

$$J = \int_0^T [(X - X^*)^T Q (X - X^*) + u^T R u] \, dt.$$

For planetary defense, X^* is not necessarily a particular orbit.

It may instead denote a safe set

$$X(t) \in \mathcal{S}_{\text{safe}}.$$

The problem then becomes one of safe-set control.

58 Viability

Define

$$\mathcal{K} = \{X : \text{no unacceptable encounter occurs}\}.$$

The viability kernel is the set of initial states from which some admissible control keeps the system in $\mathcal{K}$:

$$\text{Viab}(\mathcal{K}) = \{X_0 : \exists u(t) \text{ such that } X_t \in \mathcal{K}\}.$$

Planetary defense can therefore be formulated as maintaining strategically important bodies inside a viability domain.

59 Robust Control

Let uncertain dynamics be

$$\dot{X} = f(X, u, \theta),$$

where

$$\theta \in \Theta.$$

A robust controller solves

$$\min_u \max_{\theta \in \Theta} J(X, u, \theta).$$

This is appropriate when physical parameters, material response, or future perturbations are uncertain.

60 Strategic Reserve Mass

Not all resource-bearing mass should necessarily be extracted immediately.

Define a strategic reserve stock

$$M_t^R.$$

Its law of motion is

$$\dot{M}_t^R = -X_t.$$

Reserve value may include

$$V^R = V^{\text{economic}} + V^{\text{security}} + V^{\text{option}} + V^{\text{science}}.$$

This is analogous to strategic terrestrial reserves, but the relevant horizons may be much longer.

61 Celestial Inventory Theory

Harvested material can become orbital inventory.

Let inventory satisfy

$$\dot{I}_t = X_t - D_t - \delta I_t,$$

where X_t is production and D_t is demand.

Holding inventory has cost

$$C_H(I_t)$$

but can reduce supply risk.

Therefore

$$\min \mathbb{E} \int e^{-\rho t} [C_H(I_t) + C_S(D_t - I_t)] dt,$$

where C_S represents shortage costs.

Celestial mining therefore naturally connects to inventory economics.

62 A General Mass-Transformation Matrix

Let the mass-state vector be

$$\mathbf{m}_t = \begin{pmatrix} m_t^N \\ m_t^I \\ m_t^F \\ m_t^W \end{pmatrix},$$

representing natural, infrastructure, feedstock, and waste mass.

A transformation system is

$$\dot{\mathbf{m}} = A_t \mathbf{m}.$$

Mass conservation requires

$$\mathbf{1}^T \dot{\mathbf{m}} \approx 0$$

when relativistic mass–energy conversion is negligible.

Hence

$$\mathbf{1}^T A_t \mathbf{m} = 0.$$

This provides a compact accounting framework for strategic mass management.

63 The Mass-Management Feasibility Theorem

Theorem 63.1 (Mass-Management Feasibility). *Suppose a celestial intervention maps initial mass distribution $\mathbf{m}_0$ into terminal distribution $\mathbf{m}_T$. In a non-relativistic accounting approximation, any physically feasible transformation must satisfy*

$$\boxed{\mathbf{1}^T \mathbf{m}_T = \mathbf{1}^T \mathbf{m}_0.}$$

Therefore strategic mass management can redistribute and transform mass but cannot simply eliminate the managed mass.

Proof. The result follows directly from conservation of mass in the stated non-relativistic accounting approximation. Any extraction, transport, processing, or construction operation reallocates mass among categories. Hence the sum remains invariant. $\square$

64 Strategic Complexity Reduction

Suppose the natural environment contains

$$N$$

small objects.

After resource conversion, their material may be managed through

$$K$$

controlled infrastructure stocks, with

$$K \ll N.$$

The physical number of particles has obviously not decreased in the relevant microscopic sense. But strategic state complexity can fall:

$$\boxed{N_{\text{uncontrolled objects}} \rightarrow K_{\text{managed stocks}}.}$$

This is the correct meaning of strategic mass compression.

65 A Complexity Functional

Define strategic complexity

$$\mathcal{C} = \sum_i w_i d_i,$$

where d_i is the state dimension requiring independent tracking and w_i is its strategic importance.
A mass-management policy can reduce complexity if

$$\mathcal{C}_{\text{after}} < \mathcal{C}_{\text{before}}$$

without increasing total risk beyond tolerance.
Hence the planner may solve

$$\min_{\pi} [J_{\text{risk}}(\pi) + \lambda \mathcal{C}(\pi)].$$

66 The Strategic Mass-Management Principle

Principle 66.1 (Strategic Mass-Management Principle). *A civilization should not maximize the number of independently managed celestial bodies. It should seek the smallest dynamically adequate and welfare-enhancing strategic architecture consistent with safety, scientific preservation, physical law, and future option value.*

Symbolically,

$$\boxed{\min N_{\text{strategic}}}$$

subject to

$$\text{prediction error} \leq \epsilon,$$

$$\text{risk} \leq \bar{H},$$

$$\text{welfare} \geq \bar{W}.$$

67 Vector Graphic: Strategic Mass-Management Pipeline

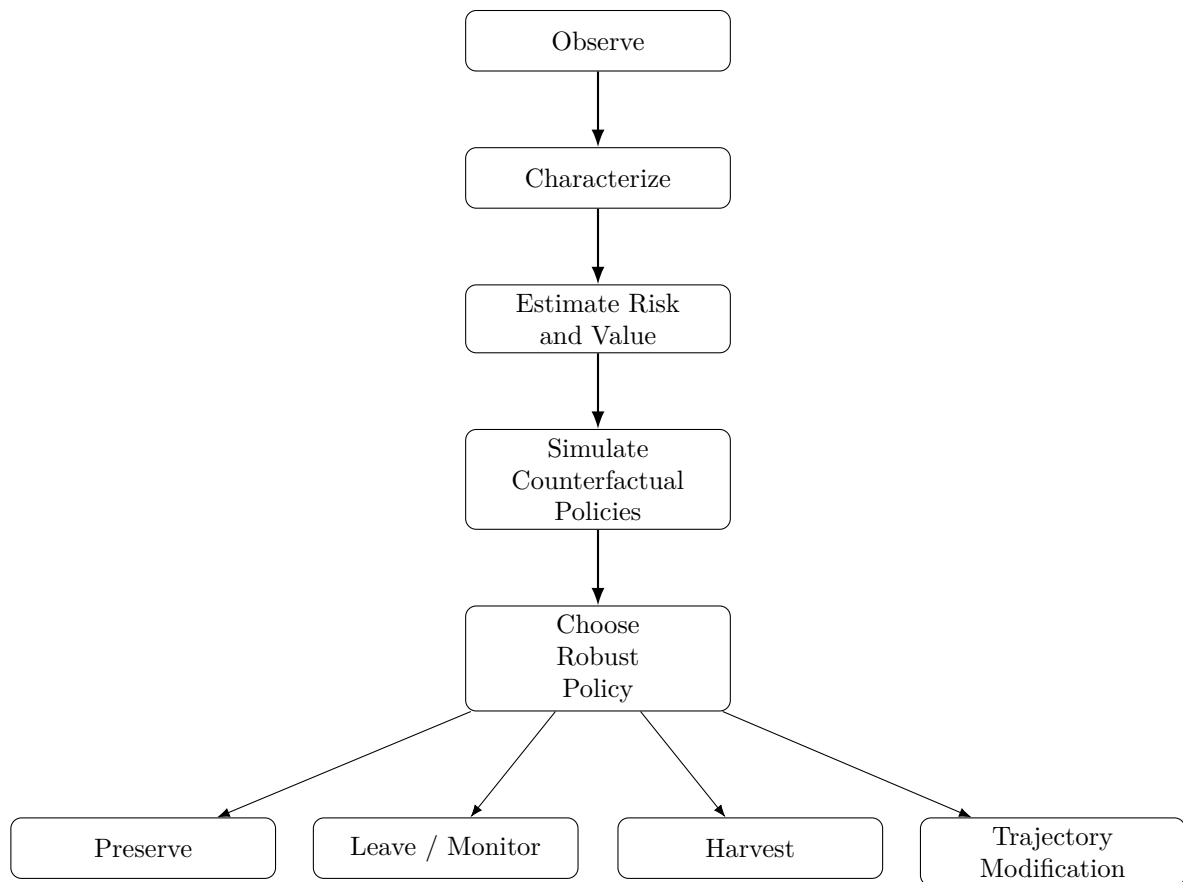


Figure 1: The Strategic Mass-Management decision pipeline.

68 Vector Graphic: Natural Mass to Managed Capital

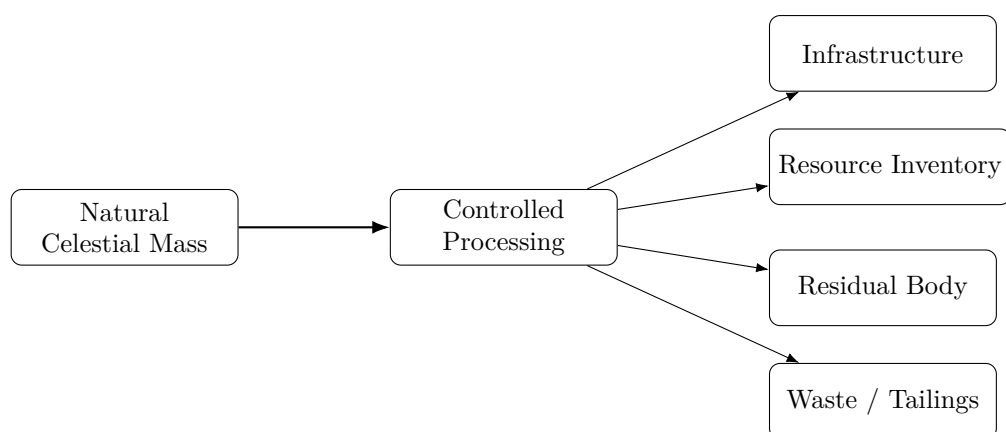


Figure 2: Harvesting transforms the distribution and economic role of mass; it does not make mass disappear.

69 Vector Graphic: Strategic Hierarchy of Celestial Mass

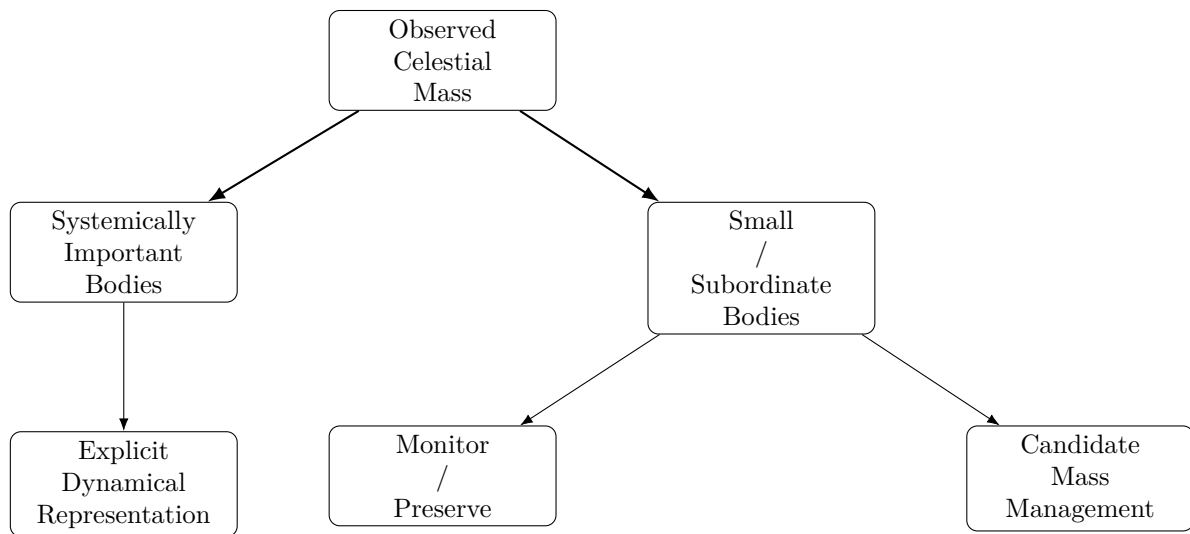


Figure 3: Mass and dynamical importance determine model treatment; discovery alone does not.

70 Vector Graphic: Risk–Resource Plane

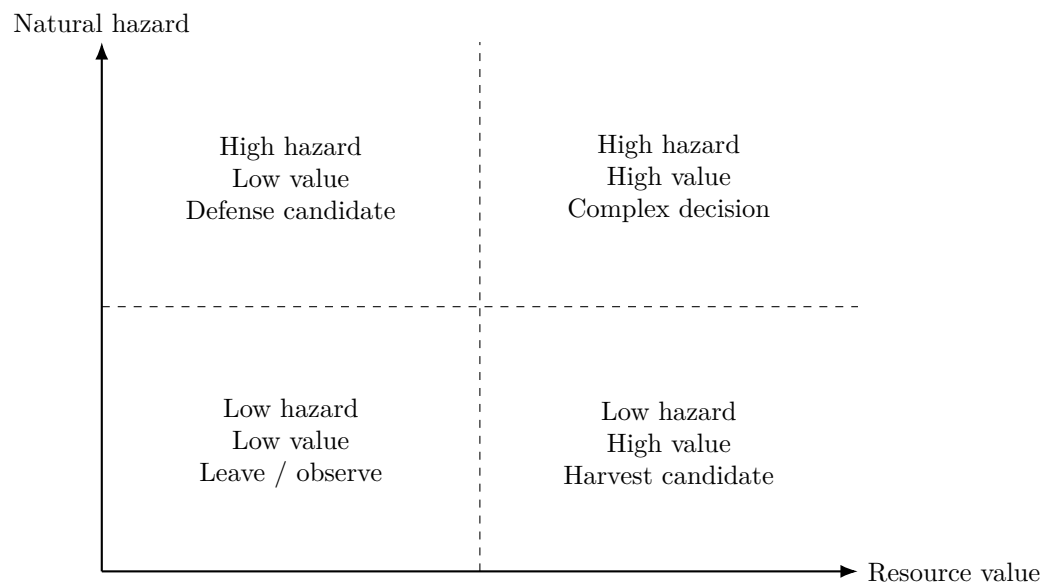


Figure 4: A simplified risk–resource classification. Intervention risk, scientific value, and uncertainty add further dimensions.

71 Vector Graphic: Feedback Architecture

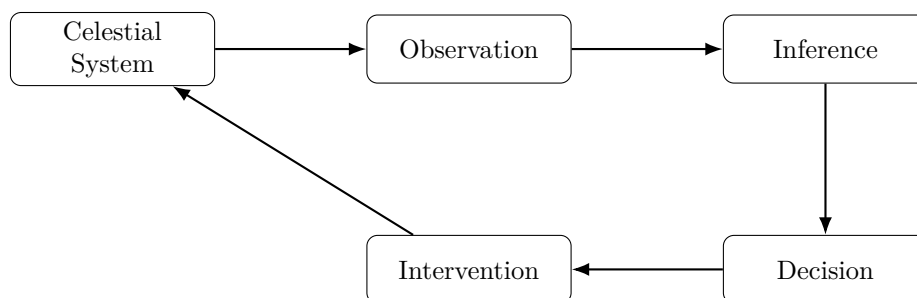


Figure 5: Strategic mass management is a closed-loop inference and control problem.

72 Policy Comparison Table

Table 1: Stylized celestial mass-management policies.

Policy	Primary objective	Main benefit	Main risk
Preserve	Scientific or natural conservation	Retains information and option value	Foregone resource value
Leave undisturbed	Avoid unnecessary intervention	Minimal intervention-generated risk	Natural hazard remains
Monitor	Improve information	Reduces uncertainty	Observation cost
Harvest	Convert natural resources into capital	Resource and infrastructure value	Orbital disturbance and mission failure
Trajectory modification	Reduce credible encounter risk	Planetary protection	Incorrect or poorly controlled intervention

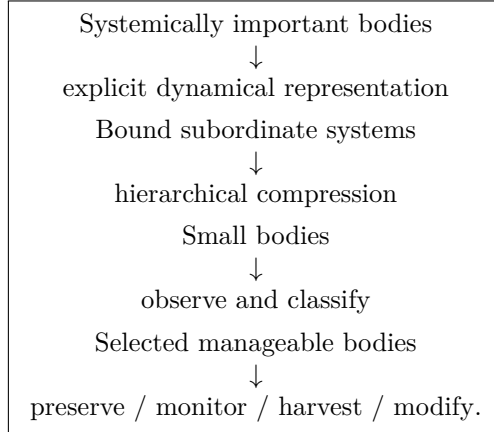
73 Comparison of Strategic Mass Types

Table 2: Conceptual classification of celestial masses.

Mass type	Typical role	Preferred model treatment	Policy question
Inhabited planet	Civilizational center	Explicit	How should it be protected?
Backup world	Redundancy	Explicit	Does it reduce correlated risk?
Major planet	Systemic gravitational body	Explicit	What perturbations does it generate?
Natural satellite	Hierarchical subsystem	Barycentric/internal	When is compression adequate?
Small asteroid	Minor natural body	Adaptive	Observe, preserve, harvest, or ignore?
Hazardous object	Risk source	High-resolution posterior	Is intervention justified?
Resource stock	Economic input	Inventory state	When should extraction occur?

74 A Hierarchical Solar-System Policy

The resulting conceptual hierarchy is



There is consequently no requirement that

$$N_{\text{model}} = N_{\text{observed}}.$$

Instead,

$N_{\text{model}} = N_{\text{dynamically and strategically necessary}}.$
--

75 Connection to the Strategic Four-Body Problem

The Strategic Four-Body Problem asks how a small collection of major bodies can be modeled dynamically and strategically.

The Strategic Mass-Management Problem adds a complementary rule:

do not enlarge the strategic state space unnecessarily.

If a newly considered mass is

small,

weakly coupled,

non-autonomous,

or

manageable as a resource or hazard,

it need not automatically become another strategic peer.

This changes the direction of research from

$$4 \rightarrow 5 \rightarrow 6 \rightarrow 7 \rightarrow \dots$$

to

identify the smallest adequate strategic architecture.
--

76 Connection to the Revolutionary Five-Body Problem

The Revolutionary Five-Body Problem established the conceptual distinction

$$N_B = 5, \quad N_{\text{eff}} = 4$$

through the Earth–Moon hierarchy.

The present framework generalizes the logic.

There are now at least three mechanisms by which physical multiplicity need not become strategic multiplicity:

hierarchical binding,
dynamical negligibility,
controlled mass transformation.

Thus the effective strategic system can remain substantially smaller than the total catalog of known celestial objects.

77 Research Program

A rigorous Strategic Mass-Management research program should proceed as follows:

1. Construct probabilistic orbital models of candidate objects.
2. Quantify object-specific hazard rather than relying on location alone.
3. Estimate mass, shape, rotation, composition, and structural uncertainty.
4. Define systemic gravitational importance.
5. Establish hierarchical compression boundaries.
6. Estimate scientific and preservation value.
7. Estimate prospective resource value.
8. Model extraction-induced momentum transfer.
9. Model fragmentation and ejecta risk.
10. Compare intervention with non-intervention.
11. Construct robust optimization models under ambiguity.
12. Include future technological learning and real-option value.
13. Incorporate intergenerational welfare.
14. Model international strategic externalities.
15. Develop transparent planetary-defense governance.
16. Construct digital twins of strategically relevant objects.
17. Develop physics-informed machine-learning models.
18. Use active learning to allocate observation resources.
19. Validate all approximations against high-fidelity celestial-mechanics simulations.
20. Promote a body to explicit strategic status only when the required accuracy, hazard, or systemic importance demands it.

78 Main Propositions

The paper can be summarized by the following propositions.

Proposition 78.1. *More observed celestial bodies do not necessarily imply more independent strategic units.*

Proposition 78.2. *Resource extraction redistributes celestial mass rather than eliminating it.*

Proposition 78.3. *Harvesting is not necessarily equivalent to hazard reduction.*

Proposition 78.4. *Fragmentation can increase rather than decrease state-space complexity and risk.*

Proposition 78.5. *Hazard depends on dynamical trajectories and probability distributions, not merely present spatial proximity.*

Proposition 78.6. *A massive gravitational body may require explicit systemic representation even when smaller bodies can be compressed, ignored, or treated as manageable resources.*

Proposition 78.7. *Observation has economic and strategic value because information can reduce uncertainty and improve subsequent decisions.*

Proposition 78.8. *When intervention is irreversible and technological progress is expected, waiting can possess positive option value.*

79 Conclusion

The Strategic Mass-Management Problem begins from a simple refusal to accept an unnecessary modeling rule:

another celestial body $\nrightarrow$ another independent strategic body.

Celestial bodies differ radically in mass, orbit, composition, scientific importance, economic value, controllability, and systemic gravitational influence.

The appropriate policy is therefore classification rather than automatic addition.

The fundamental architecture is

observe $\rightarrow$ characterize $\rightarrow$ estimate $\rightarrow$ simulate $\rightarrow$ classify $\rightarrow$ act or refrain from acting.

For any body i ,

$$a_i^* = \arg \max_{a_i \in \mathcal{A}} \mathbb{E}[W(a_i) \mid Y].$$

Possible outcomes include

preserve, leave undisturbed, monitor, harvest, modify trajectory.

The crucial insight is that harvesting is a transformation rather than annihilation:

$$m_{\text{natural}} \rightarrow m_{\text{infrastructure}} + m_{\text{inventory}} + m_{\text{waste}} + m_{\text{residual}}.$$

This transformation can convert natural celestial capital into produced capital.

But it can also create new risks.

Therefore

resource value $\neq$ net strategic value.

A complete calculation must include

natural hazard,

intervention hazard,

economic cost,

scientific value,

uncertainty,

future option value,

and

intergenerational welfare.

The framework also establishes a boundary between small manageable masses and systemically important masses.

For body j , define

$$E_j(T) = \sup_{0 \leq t \leq T} \|X_{\text{full}}(t) - X_{-j}(t)\|.$$

If

$$E_j(T) \leq \epsilon,$$

coarse representation may be acceptable.

If

$$E_j(T) > \epsilon,$$

the body requires more explicit representation.

Thus the strategic system is not determined merely by counting objects.

It is determined by

$$\boxed{\text{mass} + \text{coupling} + \text{hazard} + \text{value} + \text{information} + \text{controllability} + \text{required accuracy}.}$$

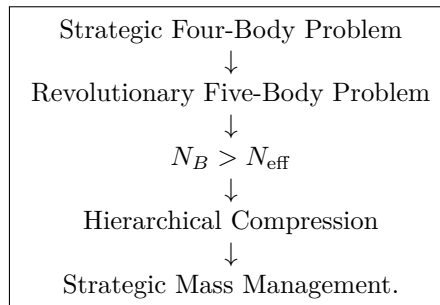
The resulting objective is not to transform every asteroid, colonize every planet, or incorporate every celestial body into an ever-growing strategic model.

It is instead

$$\boxed{\min_{\pi} \left[\frac{\text{natural}}{\text{risk}} + \frac{\text{intervention}}{\text{risk}} + \text{cost} + \frac{\text{unnecessary}}{\text{complexity}} - \frac{\text{resource}}{\text{benefit}} - \frac{\text{scientific}}{\text{benefit}} - \frac{\text{strategic}}{\text{resilience}} \right],}$$

subject to physics, uncertainty, safety, and welfare constraints.

The progression is therefore



The central question is no longer:

How many celestial bodies can be added to the strategic system?

It becomes:

Which celestial masses genuinely need to remain independent strategic objects, and which can safely remain natural, be hierarchically compressed, or eventually be transformed into useful managed capital?

That formulation turns an indefinitely expanding many-body problem into a problem of classification, dynamics, information, control, conservation, risk, and welfare.

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Glossary

Active Learning

A computational strategy that allocates observations to cases for which additional information has especially high expected value.

Asteroid Harvesting

Prospective extraction and processing of material from an asteroid. It is treated here as a future technological possibility rather than an established industrial capability.

Celestial Capital

Natural or produced physical resources located beyond Earth.

Celestial Inventory

Processed or recoverable extraterrestrial material maintained as a managed resource stock.

Celestial Mass

Matter contained in a naturally occurring or constructed astronomical object.

Digital Twin

A computational representation of a physical object continuously updated using observations.

Effective Strategic Unit

A body or subsystem that must be represented as an independent unit at the chosen strategic resolution.

Fragmentation Risk

The possibility that breaking a celestial object into multiple pieces creates additional hazardous trajectories or increases uncertainty.

Hazard Functional

A mathematical mapping from physical states and probabilities into a measure of expected harm.

Hierarchical Compression

Representation of a bound or weakly relevant subsystem through a smaller set of effective variables.

Hierarchical Compression Boundary

The boundary beyond which coarse representation violates a specified accuracy, welfare, control, or risk tolerance.

Information Capital

Accumulated observational knowledge that improves prediction and decision making.

Intervention Risk

Risk created by the attempt to alter, harvest, process, or redirect a celestial object.

Mass Management

Deliberate classification, preservation, observation, transformation, or trajectory management of celestial mass.

Mass Transformation

Conversion of natural celestial material into infrastructure, inventory, residual material, or waste while respecting conservation laws.

Natural Celestial Capital

Naturally occurring extraterrestrial resources before deliberate economic transformation.

Non-Addition Principle

The principle that discovering another celestial body does not automatically require promoting it to an independent strategic unit.

Planetary Defense

Activities concerned with detecting, characterizing, assessing, and when necessary mitigating potentially hazardous near-Earth objects.

Produced Celestial Capital

Infrastructure and productive assets constructed using extraterrestrial materials.

Real Option

The value of retaining the ability to undertake an irreversible investment at a later date.

Resource-Scale Mass

A celestial mass whose principal strategic relevance may arise from its potential resource content rather than its systemic gravitational influence.

Safe-Intervention Principle

The principle that physical intervention should occur only when its risk-adjusted expected benefit dominates non-intervention and other admissible policies.

Strategic Complexity

The amount of independently relevant state information required for prediction, control, or decision making.

Strategic Mass-Management Problem

The problem of optimally determining which celestial masses should be preserved, ignored, monitored, harvested, or subjected to trajectory management under physical, economic, informational, strategic, and welfare constraints.

Strategic Reserve Mass

Celestial material deliberately preserved for future economic, scientific, security, or option value.

Systemically Important Celestial Body

A body whose omission or excessive compression causes unacceptable errors in the dynamics or strategic analysis of a target system.

Trajectory Modification

A deliberate change to the orbit of a celestial object for a legitimate safety or management purpose.

Value of Information

The increase in expected optimal welfare generated by obtaining additional information before making a decision.

Viability Kernel

The set of initial states from which admissible controls can keep a dynamical system within a specified safe region.

The End